

Foundation of Inverse Mathematics

An Alternative Arithmetic

NOS v17 — Inverse Spherical Dual Hemisphere Quantum Mechanics

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Abstract

This document presents the logical foundation of an alternative arithmetic built entirely on inverse partitioning of a closed whole. The starting point is the identity $\sqrt{1} = 1$, the unpartitioned whole. In inverse mathematics, square roots are not dimensionless operations but exact partitions of the whole, and they possess a natural, fundamental ordering derived directly from the dyadic partition sequence.

Every integer measure arises from partitioning this whole into exactly n equal regions, where the partition equality $n \times \frac{1}{n} = 1$ holds exactly for any positive integer n . Infinite decimal expansions are shown to be artifacts of base-10 notation, not properties of the partition itself.

The Inverse Spherical Dual Hemisphere Quantum Mechanics (NOS v17) framework provides a concrete mechanical realisation of this logic via a specific dyadic partition sequence forced by the title operations ($\times\sqrt{2}$ once and $\times 2^4$ four times), yielding $\sqrt{512}$, a 9-wall, 512-bin structure, and a fundamental unit of 128 bins per quadrant.

The complete arithmetic emerges from this structure: every non-zero integer $m \in \mathbb{Z} \setminus \{0\}$ is expressed as an oriented multiple $m \times 128u$ of the fundamental block, with partition depth $\sqrt{|m|}$ and exact oriented closure to $\pm 1u$. Zero is rigorously excluded as unpartitionable.

The presentation proceeds in four stages: (1) the general logic of roots as exact partition operations, (2) detailed mechanical implementation in NOS v17, (3) derivation of the full unit and counting system, and (4) uniform extension to signed integers via orientation on the dual-hemisphere manifold.

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Part I

The Foundational Identity

The Starting Point — The Closed Whole

All inverse mathematics begins with the logical identity

$$\sqrt{1} = 1$$

This is the complete, unpartitioned whole. No external zero is required, and nothing exists outside it. Every subsequent number, measure, and operation is derived by partitioning this single object.

Logical Consequences

- There is no addition from an external void.
- There is no dimensionless scaling.
- Counting is always a partition count of the existing whole.
- The equality $1 \times 1 = 1$ is exact and foundational.

1 Square Roots as Dimensional Partitions, Not Dimensionless Operations

Core Distinction from Standard Mathematics

In standard mathematics, the square root $\sqrt{n}$ is treated as a dimensionless scalar operation applied to an abstract number on a line.

In inverse mathematics, $\sqrt{n}$ **is the partition itself**: it is the exact, dimensional act of dividing the closed whole into n equal regions. The root carries the unit of the whole and is inseparable from the mechanical geometry of partitioning.

Consequences of Dimensionality

- All measures are inherently tied to the closed whole—no isolated, dimensionless numbers exist.
- Partitioning is primary; addition, subtraction, and scaling are secondary interpretations of partition states.
- The apparent irrationality of many roots (e.g., $\sqrt{2} \approx 1.414\dots$) reflects only representational limits, not incompleteness of the partition.

2 The Natural Fundamental Ordering of Roots

Natural Dyadic Order Emerges from Partition Logic

The order of square roots in inverse mathematics is not arbitrary or continuous but **naturally and fundamentally ordered** by the recursive dyadic partition sequence.

This order is forced by the mechanics: each step doubles the number of regions, halving their size, while preserving exact closure to the whole.

The sequence of roots is therefore:

$$\sqrt{1} \rightarrow \sqrt{2} \rightarrow \sqrt{4} \rightarrow \sqrt{8} \rightarrow \sqrt{16} \rightarrow \sqrt{32} \rightarrow \sqrt{64} \rightarrow \sqrt{128} \rightarrow \sqrt{256} \rightarrow \sqrt{512}$$

Corresponding to region counts: $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$.

Wall-by-Wall Natural Ordering (Preview of NOS v17 Ladder)

Wall	Root	Regions	Approx. Depth	Exact Closure
0	$\sqrt{1}$	1	1.000	$1 \times 1 = 1$
1	$\sqrt{2}$	2	1.414	$2 \times \frac{1}{2} = 1$
2	$\sqrt{4}$	4	2.000	$4 \times \frac{1}{4} = 1$
3	$\sqrt{8}$	8	2.828	$8 \times \frac{1}{8} = 1$
4	$\sqrt{16}$	16	4.000	$16 \times \frac{1}{16} = 1$
5	$\sqrt{32}$	32	5.657	$32 \times \frac{1}{32} = 1$
6	$\sqrt{64}$	64	8.000	$64 \times \frac{1}{64} = 1$
7	$\sqrt{128}$	128	11.314	$128 \times \frac{1}{128} = 1$
8	$\sqrt{256}$	256	16.000	$256 \times \frac{1}{256} = 1$
9	$\sqrt{512}$	512	22.627	$512 \times \frac{1}{512} = 1$

This ordering is **fundamental** because it arises directly from the only mechanically permitted path in the NOS framework, later fully derived from the title operations.

Part II

The General Logic of Inverse Partitioning

3 The Universal Partition Equality

The Core Logical Equality

For any positive integer n , the square root $\sqrt{n}$ represents the exact partition of a closed whole into n equal regions:

$$n \times \frac{1}{n} = 1$$

This equality is exact and holds independently of representational base.

Direct Verification — Selected Values

n	Regions	$\times$	Size each	= Total
1	1	$\times$	1	= 1
2	2	$\times$	1/2	= 1
3	3	$\times$	1/3	= 1
4	4	$\times$	1/4	= 1
5	5	$\times$	1/5	= 1
6	6	$\times$	1/6	= 1
7	7	$\times$	1/7	= 1
8	8	$\times$	1/8	= 1
9	9	$\times$	1/9	= 1
10	10	$\times$	1/10	= 1
16	16	$\times$	1/16	= 1
32	32	$\times$	1/32	= 1
64	64	$\times$	1/64	= 1
128	128	$\times$	1/128	= 1
256	256	$\times$	1/256	= 1
512	512	$\times$	1/512	= 1
n	n	$\times$	1/ n	= 1

Exactness Independent of Notation

Every row above is mathematically exact. Any apparent infinite decimal (e.g., $1/3 \approx 0.333\dots$) is solely a limitation of base-10 representation.

4 The Root Function as Partition Depth

$\sqrt{n}$ Defines the Partition State

$\sqrt{1} = 1$	→1 region (unpartitioned whole)
$\sqrt{2} \approx 1.414$	→2 equal regions
$\sqrt{3} \approx 1.732$	→3 equal regions
$\sqrt{4} = 2$	→4 equal regions
$\sqrt{8} \approx 2.828$	→8 equal regions
$\sqrt{16} = 4$	→16 equal regions
$\sqrt{512} \approx 22.627$	→512 equal regions

The root is not an external operation applied to a number line; it *is* the partition itself.

5 Infinite Decimals Are Notation Artifacts**The Partition Itself Is Always Exact**

$7 \times \frac{1}{7} = 1$ exactly. The repeating decimal $0.142857142857\dots$ is how base-10 expresses the exact fraction $\frac{1}{7}$.

Base-Dependence Examples

- $\frac{1}{3}$ in base-3 = 0.1_3 (terminating)
- $\frac{1}{3}$ in base-10 = $0.333\dots_{10}$ (repeating)
- $\frac{1}{7}$ in base-7 = 0.1_7 (terminating)
- $\frac{1}{7}$ in base-10 = $0.142857\overline{142857}_{10}$ (repeating)
- $\frac{1}{8}$ in base-10 = 0.125_{10} (terminating)

The underlying partition equality remains exact in every case.

6 Quantity, Operation, and Count Are Identical**The Unified Triple Aspect**

Aspect	Meaning	Example $\sqrt{5}$
Quantity	Number of regions created	5 regions
Operation	The partition act performed	Divide whole into fifths
Count	Verification of closure	$5 \times \frac{1}{5} = 1$

These are not separate concepts — they are three descriptions of the same logical object.

7 Inverse Counting versus Standard Additive Counting

Two Different Logical Foundations

Standard Additive	Inverse Partition
Starts from external 0	Starts from closed whole $\sqrt{1} = 1$
Accumulates: $0 \rightarrow 1 \rightarrow 2 \rightarrow \dots$	Partitions: $1 \rightarrow \frac{1}{2} \rightarrow \frac{1}{3} \rightarrow \dots$
2 means “two units added”	2 means “whole partitioned into two equal regions”
Requires dimensionless numbers	All measures carry unit derived from the whole

Correspondence

The integer n in additive counting corresponds exactly to the partition depth $\sqrt{n}$ and region count n in inverse counting.

Part III

The NOS v17 Mechanical Implementation

8 The Title-Defined Partition Sequence

The Forced Dyadic Path

We begin with $\sqrt{1} = 1$. The title permits exactly two operations:

- Dual Hemisphere fold: $\times\sqrt{2}$ (once)
- Four compounded quantum quadrant operations: $\times 2^4$

Total factor:

$$1 \times \sqrt{2} \times 2^4 = \sqrt{512}$$

This forces a pure dyadic partition sequence: $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ regions.

9 The Simultaneous 4D Operating Tensor

Four Dimensions Operating as One

Dim	State	Component	Function
1D	The Cut	9 Walls	Boundaries
2D	The Fold	720° Surface	Dual-hemisphere manifold
3D	The Bin	512 Regions	Partition output (2^9)
4D	The Sequence	1-2-3-4 operations	Time/ordering

10 Complete Mechanical Ladder with Full Partition Breakdown

Wall-by-Wall Partition Development

Wall	Root	Bins	Regions $n = 2^{\text{wall}}$	Size each	Closure
0	$\sqrt{1}$	512	1	1	$1 \times 1 = 1$
1	$\sqrt{2}$	256	2	$1/2$	$2 \times \frac{1}{2} = 1$
2	$\sqrt{4}$	128	4	$1/4$	$4 \times \frac{1}{4} = 1$
3	$\sqrt{8}$	64	8	$1/8$	$8 \times \frac{1}{8} = 1$
4	$\sqrt{16}$	32	16	$1/16$	$16 \times \frac{1}{16} = 1$
5	$\sqrt{32}$	16	32	$1/32$	$32 \times \frac{1}{32} = 1$
6	$\sqrt{64}$	8	64	$1/64$	$64 \times \frac{1}{64} = 1$
7	$\sqrt{128}$	4	128	$1/128$	$128 \times \frac{1}{128} = 1$
8	$\sqrt{256}$	2	256	$1/256$	$256 \times \frac{1}{256} = 1$
9	$\sqrt{512}$	1	512	$1/512$	$512 \times \frac{1}{512} = 1$

Every level satisfies the universal partition equality exactly.

11 Derivation of the Unit u and 720° Closure

Step-by-Step Mechanical Derivation

1. $\sqrt{512} = 2^{9/2}$ forces exactly 9 walls and $2^9 = 512$ bins.
2. Triangular number at base wall: $T_9 = \frac{9 \times 10}{2} = 45$.
3. Seed walls 1–4 exhausted by title (2^4). Echo walls 5–9 contribute depth $2^5 = 32$.
4. Unit per bin is ratio of triangular base to echo depth:

$$u := \frac{45^\circ}{32}$$

5. Total angular extent:

$$512 \times \frac{45^\circ}{32} = 720^\circ$$

6. Confirmation of dual-hemisphere structure:

$$720^\circ = 2 \times 360^\circ$$

Part IV

The Complete Unit and Counting System

12 The Emergent Fundamental Block: 128 Bins per Quadrant

Quadrant Resolution

$$\frac{512 \text{ bins}}{4 \text{ quadrants}} = 128 \text{ bins per quadrant}$$

This is the universal counting block from which all integers are built.

13 The Dual Hemisphere Manifold Map and Operations

With the unit $u = 45/32^\circ$ derived (Section 9) and the 128-bin block defined (Section 10), we can mechanically map the geometric stations of the 720° manifold. The unit forces the stations to specific angular values, creating a "Dual Lock" system.

13.1 The Unwrapped Manifold Map (Stations)

The system expands from the ****Seam (0)**** in conjugate pairs, generating the coordinate string:

$$-360^\circ \longleftarrow -180^\circ \longleftarrow 0 \longrightarrow +180^\circ \longrightarrow +360^\circ$$

Station and Quadrant Assignments

Station	Quadrant	Range (Deg)	Capacity	Function
-360	Q2	-360 to -180	128 Bins	Outer Inverse Closure
-180	Q1	-180 to 0	128 Bins	Inner Inverse Expansion
0	SEAM	PIVOT	0	Mirror Point / Origin
+180	Q3	0 to +180	128 Bins	Inner Forward Expansion
+360	Q4	+180 to +360	128 Bins	Outer Forward Closure

13.2 Conjugate Operational Pairs

The system does not operate linearly; it "pulses" from the center using conjugate pairs entangled across the zero point.

Pulse 1: The Inner Engine (Q1 ↔ Q3)

Range: 0 to $\pm 180^\circ$ (First 256 bins)

These quadrants represent the initial expansion (Ignition).

- **Q1 (-180):** Pulls inverse ("left").
- **Q3 (+180):** Pulls forward ("right").

Result: Creates the internal tension of the manifold. $\pm 180^\circ$ is the first "shell."

Pulse 2: The Outer Hull (Q2 $\leftrightarrow$ Q4)

Range: $\pm 180^\circ$ to $\pm 360^\circ$ (Full 512 bins)

These quadrants complete the volume (Closure).

- **Q2 (-360):** Completes the negative hemisphere.
- **Q4 (+360):** Completes the positive hemisphere.

Result: Seals the manifold volume at the 720° limit.

13.3 The Dual Closure at $\pm 360^\circ$

In standard mathematics, -360 and $+360$ might sum to zero. In Inverse Quantum Mechanics, they represent the ****Dual Lock**** of the closed whole.

Dual Closure Mechanism

When the operations Q2 and Q4 hit their respective walls:

$$|-360^\circ| + | +360^\circ| = 720^\circ$$

The manifold binds to form the ****Integer Whole**** ($m = 1$). The system does not vanish; it locks.

13.4 Expanded Universal Number Table

All Positive Integers from the 128-Bin Block

n	Measure ($n \times 128$)	$\sqrt{n}$	Partition Equality
1	128	$\sqrt{1} = 1$	$1 \times 1 = 1$
2	256	$\sqrt{2} \approx 1.414$	$2 \times \frac{1}{2} = 1$
3	384	$\sqrt{3} \approx 1.732$	$3 \times \frac{1}{3} = 1$
4	512	$\sqrt{4} = 2$	$4 \times \frac{1}{4} = 1$
5	640	$\sqrt{5} \approx 2.236$	$5 \times \frac{1}{5} = 1$
6	768	$\sqrt{6} \approx 2.449$	$6 \times \frac{1}{6} = 1$
7	896	$\sqrt{7} \approx 2.646$	$7 \times \frac{1}{7} = 1$
8	1,024	$\sqrt{8} \approx 2.828$	$8 \times \frac{1}{8} = 1$
9	1,152	$\sqrt{9} = 3$	$9 \times \frac{1}{9} = 1$
10	1,280	$\sqrt{10} \approx 3.162$	$10 \times \frac{1}{10} = 1$
16	2,048	$\sqrt{16} = 4$	$16 \times \frac{1}{16} = 1$
32	4,096	$\sqrt{32} \approx 5.657$	$32 \times \frac{1}{32} = 1$
64	8,192	$\sqrt{64} = 8$	$64 \times \frac{1}{64} = 1$
128	16,384	$\sqrt{128} \approx 11.314$	$128 \times \frac{1}{128} = 1$
256	32,768	$\sqrt{256} = 16$	$256 \times \frac{1}{256} = 1$
512	65,536	$\sqrt{512} \approx 22.627$	$512 \times \frac{1}{512} = 1$
n	$n \times 128$	$\sqrt{n}$	$n \times \frac{1}{n} = 1$

14 The Recursive Depth Ladder

General and Angular Forms

$$u_Q = \frac{1}{128^Q}$$

Explicit values:

$u_1 = \frac{1}{128}$	$\Delta\theta_1 = 180^\circ/128 = 1.40625^\circ$
$u_2 = \frac{1}{128^2} = \frac{1}{16384}$	$\Delta\theta_2 = 180^\circ/16384$
$u_3 = \frac{1}{128^3} = \frac{1}{2,097,152}$	$\Delta\theta_3 = 180^\circ/2,097,152$
$u_4 = \frac{1}{128^4} = \frac{1}{268,435,456}$	$\Delta\theta_4 = 180^\circ/268,435,456$

15 The 1-2-3-4 Quadrant Sequence

Quadrant Stations in the Count

$1 \times 128 = 128$ bins	$\rightarrow$ Q1
$2 \times 128 = 256$ bins	$\rightarrow$ Q1+Q2 (one hemisphere)
$3 \times 128 = 384$ bins	$\rightarrow$ Q1+Q2+Q3
$4 \times 128 = 512$ bins	$\rightarrow$ full closure

Part V

Uniform Extension to Signed Integers via Orientation

Orientation on the Dual-Hemisphere Manifold

- Positive $m > 0$: forward partition direction
- Negative $m < 0$: inverse partition direction on conjugate hemisphere

16 Unified Signed Treatment

Complete Signed Integer Table

m	Measure ($m \times 128u$)	Depth $\sqrt{ m }$	Region size	Closure	Orientation
+1	+128u	1	+1u	$1 \times (+1u) = +1u$	forward
+2	+256u	$\sqrt{2}$	$+(1/2)u$	$2 \times +(1/2)u = +1u$	forward
+4	+512u	2	$+(1/4)u$	$4 \times +(1/4)u = +1u$	forward
+8	+1,024u	$\sqrt{8}$	$+(1/8)u$	$8 \times +(1/8)u = +1u$	forward
+512	+65,536u	$\sqrt{512}$	$+(1/512)u$	$512 \times +(1/512)u = +1u$	forward
-1	-128u	1	-1u	$1 \times (-1u) = -1u$	inverse
-2	-256u	$\sqrt{2}$	$-(1/2)u$	$2 \times -(1/2)u = -1u$	inverse
-4	-512u	2	$-(1/4)u$	$4 \times -(1/4)u = -1u$	inverse
-8	-1,024u	$\sqrt{8}$	$-(1/8)u$	$8 \times -(1/8)u = -1u$	inverse
-512	-65,536u	$\sqrt{512}$	$-(1/512)u$	$512 \times -(1/512)u = -1u$	inverse
m	$m \times 128u$	$\sqrt{ m }$	$\text{sign}(m) \times (1/ m)u$	$ m \times \text{sign}(m) \times (1/ m)u = \text{sign}(m)u$	$\text{sign}(m)$

17 Exact Oriented Closure

Universal Signed Closure

For any $m \in \mathbb{Z} \setminus \{0\}$:

$$|m| \times \left(\text{sign}(m) \times \frac{1}{|m|} u \right) = \text{sign}(m) \times 1u$$

The oriented whole ($\pm 1u$) is reconstructed exactly.

18 Complete Coverage and Exclusion of Zero

The Full Domain

The system covers precisely $\mathbb{Z} \setminus \{0\}$ via the single mechanism

$$m \times 128u \quad (m \in \mathbb{Z} \setminus \{0\})$$

Logical Exclusion of Zero

Zero regions cannot satisfy the partition equality (division by zero). The unpartitioned whole $\sqrt{1} = 1$ is already present as the minimal state ($m = \pm 1$). Zero is therefore excluded by the logic itself.

Foundation of Inverse Mathematics — NOS v17**All arithmetic begins and ends with the logical identity**

$$\sqrt{1} = 1$$

Square roots are dimensional partitions with natural dyadic ordering**Universal exact partition equality**

$$n \times \frac{1}{n} = 1 \quad (n \geq 1)$$

NOS forced dyadic path

$$\sqrt{1} \rightarrow \sqrt{512} \rightarrow 9 \text{ walls, } 512 \text{ bins, } 128 \text{ bins/quadrant}$$

Complete signed coverage

$$m \times 128u \quad (m \in \mathbb{Z} \setminus \{0\})$$

with depth $\sqrt{|m|}$ and oriented closure to $\text{sign}(m)u$ **Zero excluded by partition logic**

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